In a single .java file, if we write only one public class and multiple non-public classes then java compiler will generate how many .class files?

Whether the class is public OR non-public, Java compiler will generate the .class file for all the java classes.
[Mantra: I have 10 java classes either public OR non-public so 10 .class files will be generated by compiler]

Example :

```
//Test.java [File Name]

public class Test
{

}

class Alpha
{
}

class Beta
{
}
```

→ Total 3 .class files are created i.e. Test.class, Alpha.class and Beta.class

Structure of a Java class?

1) package statement should be in the first line
2) import statement should be in the 2nd line
3) class body should be in the 3rd line

Example :

```
package com.ravi.basic;

import java.util.Arrays;

public class Test
{
     Arrays a = null;
}
```

Working with Static Method with different return type using BLC & ELC:

* As we know we can directly call a static method with the help of class name, object is not required.

Writing a java program with BLC & ELC class approach :

* It is **highly recommended** to write java programs with BLC & ELC class approach. [We should never write everything (logic + execution) in a single program as shown below]

```
Main.java
----------
public class Main
{
   void main()
   {
       doSum(12,12);
   }

   public void doSum(int x, int y)
   {
       IO.println("Sum is :"+(x+y));
   }
}
```

* If we write the java programs with the help BLC & ELC approach then BLC classes can be re-usable into multiple packages providing loose coupling as shown below :

```
//BLC
Addition.java
----------------
public class Addition
{
     public static void doSum(int x, int y)
     {
         IO.println("Sum is :"+(x+y));
     }
}
```

```
//ELC
Main.java
-----------
public class Main
{
     void main()
     {
       Addition.doSum(12,90);
     }
}
```

* BLC & ELC PROGRAM STYLE APPROACH WE CAN WRITE BY USING FOLLOWING 2 APPROACHES :

1) Single File Approach :

```
Main.java
-----------
//ELC
public class Main
{
     void main()
     {
         Addition.doSum(12,90);
     }
}

//BLC
class Addition
{
     public static void doSum(int x, int y)
     {
         IO.println("Addition of two number  is :"+(x+y));
     }
}
```

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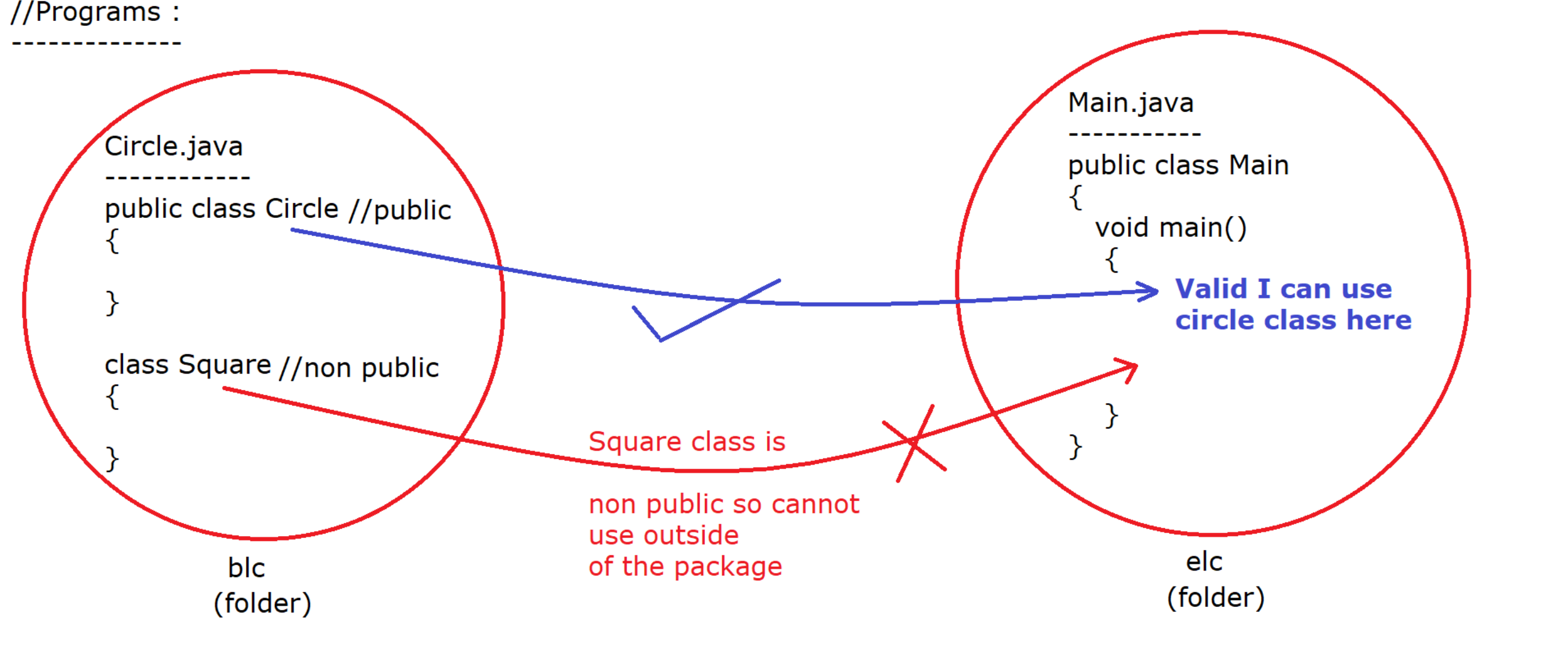
2) Multiple file approach :

```
//BLC
Addition.java
----------------
public class Addition
{
     public static void doSum(int x, int y)
     {
         IO.println("Sum is :"+(x+y));
     }
}

//ELC
public class Main
{
     void main()
     {
       Addition.doSum(10,90);
     }
}
```

//Programs :



* If we want to use the BLC class from one **package to another package** then we need to import the BLC class from current package (folder) to another package (folder) by using **import statement** but the BLC class **must be public**

* **If BLC class is non public then we cannot import to another package (folder)**

Circle.java

```
package com.ravi.blc;

public class Circle //public
{
  public static void getAreaOfCircle(double radius)
  {
      final double PI = 3.14;
      double area = PI * radius * radius;
      System.out.printf("Area of Circle is : %.2f ",area);
  }
}
```

```
package com.ravi.elc;

import com.ravi.blc.Circle;

public class Main
{
   void main()
   {
      double radius = Double.parseDouble(IO.readln("Enter the radius :"));
      Circle.getAreaOfCircle(radius);

   }
}
```

The above program is available with multiple file approach

```
package com.ravi.blc;

//BLC
public class ArithmeticOperation
{
   public static  int sum(int x, int y)
   {
       return x+y;
   }

   public static int sub(int x, int y)
   {
       return x-y;
   }

   public static int mul(int x, int y)
   {
       return x*y;
   }

   public static int div(int x, int y)
   {
       if(y==0)
       {
           System.err.println("Cannot divide a number by zero");
           return 0;
       }

       return x/y;
   }
}
```

```
package com.ravi.elc;

import com.ravi.blc.ArithmeticOperation;

public class Main
{
   void main()
   {
      int result = ArithmeticOperation.sum(12, 12);
      IO.println("Sum is :"+result);

      IO.println("Sub is :"+ArithmeticOperation.sub(200, 100));
      IO.println("Mul is :"+ArithmeticOperation.mul(10, 20));
      IO.println("Div is :"+ArithmeticOperation.div(200, 10));


   }
}
```

Working with method return type as a String :

* If a method return type is String then we can append multiple values in a single statement by using concatenation operator.

```
package com.ravi.blc;

public class Employee
{
  public static String getEmployeeDetails(int id, String name, double salary)
  {
      return "[Employee id is :"+id+", name is :"+name+", salary is :"+salary+"]";
  }
}
```

```
package com.ravi.elc;

import com.ravi.blc.Employee;

public class Main
{
   void main(String [] args)
   {
     int id = Integer.parseInt(args[0]);
     String name = args[1];
     double salary = Double.parseDouble(args[2]);

     String employeeDetails = Employee.getEmployeeDetails(id, name, salary);
     IO.println("Employee Details :");
     IO.println(employeeDetails);

   }
}
```

How to execute command line argument program in Eclipse IDE :

Right click on the program -> Run as -> Run Configuration -> Verify your project name & Main class name (ELC class) -> Provide appropriate value to the argument tab -> run the program